Chapter 1
Charge = Coulombs (integral of current, area under the current curve)
Current = Coulombs/Second (derivative of Charge, slope of charge, change in charge over time, dq/dt)
Voltage = Joules/Coulomb
Power: P=IV, P=(V)^2/R, P=R(I)^2
Amp-hours = Current * time
Watt-hours (Energy) = Power*time
Watt-hours * $$$ = cost due to electricity

Chapter 2
Resistance = ρ(l/A) p = resistivity constant, l=length, A = cross sectional area
Ohm's Law: V=IR, I=V/R, R=V/I
Short Circuit = 0 ohms (Resistance), any current can flow through a short
Open Circuit = ∞ ohms (Resistance), no current can flow through an open circuit
Branches, nodes, loops: B = L + N - 1
Series: Two Elements share a single node. Same **Current** for all resistors in series, different Voltages (Voltage Division).
Parallel: Two Elements connected to same two nodes. Same **Voltage** for all resistors in parallel, different Currents (Current Division).
Linearity: V/I = R creates a perfectly linear relationship for certain circuits.
Kirchhoff's Current Law (KCL): Current in=Current out
Kirchhoff's Voltage Law (KVL): The sum of Voltages around a loop = 0

Voltage Division (Series Resistors):

Two Resistors are considered in series if they have the same current pass through them. If we have a Vs and two resistors in series, here is the equations:

- V1 = i(R1)
- V2 = i(R2)
- -V + V1 + V2 = 0
- -V + i(R1) + i(R2) = 0

The current through all resistors in series is the same, so using Ohm's law:

- V1 = [(R1)/(R1 + R2)]*V
- V2 = [(R2)/(R1 + R2)]*V
- V = V1 + V2 = i(R1 + R2)

Current Division (Parallel Resistors):

Current Division

- Given the current entering the node, the voltage drop across the equivalent resistance will be the same as that for the individual resistors

$$v = iR_{eq} = \frac{iR_1R_2}{R_1 + R_2}$$

- This can be used in combination with Ohm's law to get the current through each resistor:

$$i_1 = \frac{iR_2}{R_1 + R_2} \quad i_2 = \frac{iR_1}{R_1 + R_2}$$

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Delta(Δ)-Wye(Y) Transformations:

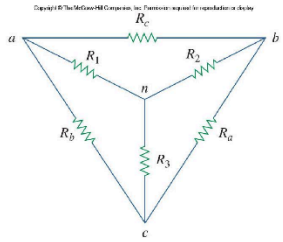
Delta to Wye

- The conversion formula for a delta to wye transformation are:

$$R_1 = \frac{R_bR_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_cR_a}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_aR_b}{R_a + R_b + R_c}$$



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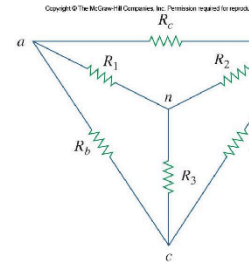
Wye to Delta

- The conversion formula for a wye to d transformation are:

$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}$$

$$R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2}$$

$$R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3}$$



Chapter 3

Nodal Analysis: Focuses on **current** flowing into and out of each node using KCL. Because V=I/R, we are actually going to find the node voltages in the end (v1, v2, v3, etc.).

Steps for Nodal Analysis:

1. Identify nodes in the circuit (use coloring method if necessary)
2. Apply KCL at each node (except for the ground node)
3. Solve the KCL equations using a matrix to find the unknown node voltages.

Tips:

- IF there is a voltage source between two nonreference nodes (v1 and v2), that becomes a **supernode.** Treat the supernode as 1 node, and write a constraint equation using KCL (current coming in = current flowing out, AND **voltage amount = v1-v2**, using the + on the Vs as the positive node voltage and - on the Vs as the negative node voltage).
- If there is a voltage source (Vs) between a nonreference node (V1) and a reference node (Ground), do [(Vs-V1)/resistance]
- If there is a Voltage Source (Vs) right next to a nonreference node (v1) and it looks like it is the same voltage, it probably is!

Do Nodal Analysis If the circuit contains:

- Many elements in parallel
- Current Sources
- Supernodes
- Circuits with fewer Nodes than meshes
- If the Node Voltage is what is being solved for

Non-Planar Circuits

Mesh Analysis: Uses KVL to find unknown "Mesh Currents" applies in planar circuits).

Steps for Mesh Analysis:

1. Assign mesh currents to meshes
2. Apply KVL to each of the meshes (loops)
3. Solve the resulting equations to get the mesh curren

Tips:

- The mesh currents use KVL calculated **CLOCKWISE** ar mesh/loop.
- If the mesh current goes **WITH** the Voltage sorce (fro -->), then the voltage source should be negative in th equation.
- If the mesh current goes **AGAINST** the Voltage sourc (+ -) -->), then the voltage source should be positive equation.
- IF there is a **resistor bordering** two meshes, make th current you are working on positive and the borderir current negative.
- If there is a **current source isolated** to a mesh, that r current = the current source.
- If there is a **current source bordering** two meshes, th classified as a **SUPERMESH**. For the constraint equat mesh currents bordering the current source and thei to determine which is positive and which is negative current is going the same direction as the current so it is positive in the constraint equation. If a mesh cur going against the current source, then it is negative i constraint equation.
- After the constraint equation is written for the SUPE you then REMOVE (create an open circuit) the wire c the current source and resistor it is attached to. Ther on the remaining Mesh, using I1 and I2 accordingly (resistors applied to I1 before the supermesh branch removed, still applies to I1, and vice versa for I2).

Do Mesh Analysis if the circuit contains:

- Many elements in series
- Voltage sources
- Supermeshes
- A circuit with fewer meshes than nodes
- If a branch/mesh current is what is being solved fo

Chapter 4

Linearity: As Voltage goes up, Current goes up proportionally. The response of a circuit to a **sum of sources** will be the **sum of the individual responses** from **each source separately**.

Linearity:

- V=iR
- k(iR) = k(V)
- V= (i1 + i2)R = (i1)R + (i2)R = V1 + V2

Superposition: If there are **two or more independent sources** there are **Thr** solve for the circuit parameters: Nodal Analysis, Mesh Analysis, Superositio

- Superposition Principle states that the voltage/Current through an el linear circuit is the **total sum of the voltages/currents** through that e to each independent source acting alone.

Steps for Superposition:

1. Turn off all independent sources except one source. Find the output current) due to that active source using techniques in Chapter 2/3
2. Repeat step 1 for each of the other independent sources

	3. Find the total contribution (voltage or current) by adding all the contr (voltages or currents) due to each independent source. Tips for Superposition: ● To remove a current source, replace it with an open circuit (i=0). ● To remove a voltage source, replace it with a short circuit (v=0).

Source Transformation: Replacing a Voltage source (Vs) in series with a resistor R by a current source (Is) in parallel with a resistor R, or vice versa. ● Vs = Is*R ● Is = Vs/R ● The current source is directed towards the positive terminal of the voltage source ● Source Transformation is not possible when R=0 or R=∞	**Thevenin's Theorem (Thevenin Equivalents):** When you have one variable element that you are trying to analyze, called the load. ● Thevenin Voltage (Vth/Voc)= Open Circuit Voltage (R=infinite) ● Thevenin Resistance (Rth)= Resistance looking into terminals a and b with **all independent sources turned off.** ● Thevenin Current (Ith/In/Isc) = Vth/Rth = Ith ● Use a voltage source in series with a resistance to replace complicated linear circuits It is possible for the result of this analysis to end up with a negative resistance. This implies the circuit is supplying power. This is reasonable with dependent sources
Norton's Theorem (Norton Equivalents): Similar to Thevenin's theorem, Norton's theorem states that a linear two terminal circuit may be replaced with an equivalent circuit containing a resistor and a current source. ● Norton Current (In)= Short Circuit Current (Isc) ● Norton Resistance (Rn) = Thevenin Resistance (Rth) ● Norton Voltage (Vn) = [Vn = In * Rn] ● The Norton Current (In) is found by short circuiting the circuit's terminals and measuring the resulting current. ● Norton current is related to Thevenin voltage and resistance by source transformation: In = Vth/Rth	**Norton vs. Thevenin** • With $\mathbf{V_{TH}}$, $\mathbf{I_N}$, and $(\mathbf{R_{TH}} = \mathbf{R_N})$ related, finding the Thevenin or Norton equivalent circuit requires that we find: • The open-circuit voltage across terminals *a and b.* • The short-circuit current at terminals *a and b.* • The equivalent or input resistance at terminals *a and b when* all independent sources are turned off.

Maximum Power Transfer:

P = [(Vth)^2/4(Rth)]

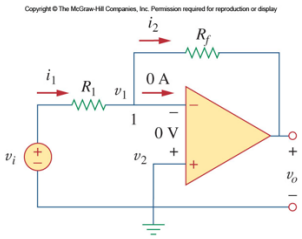
Chapter 5 Concepts

- 5 Terminals found on all Op-amps:
  a. Inverting input
  b. Non-inverting input
  c. Output
  d. Positive and negative power supplies
- Output voltage of an op-amp can be found using the following equation: Vo = A*Vd = A*(V^+-V^-)
  - A is the Open Loop Gain, which is different from the closed loop gain.
- Feedback: The output of the op-amp is fed back to the inverting terminal, giving the op-amp "Negative Feedback".
- Voltage Saturation: The output voltage of the op-amp cannot exceed the input voltages. Therefore, when an output voltage should exceed the possible voltage range, the output remains at either the minimum or maximum supply voltage.
- **Ideal Op-Amps:**
  - Infinite Open Loop Gain (A)
  - Infinite Input Resistance (Ri)
  - Zero output Resistance (Ro)
  - Zero input current to the inverting/noninverting terminals (Io)
  - Output current is NOT zero

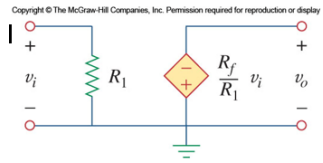
Input voltages on the inverting/noninverting terminals are equal

Inverting Op-amp (ideal)

- Ideal Op-Amp Rules apply
  - Vo = (-Rf/R1)*Vi

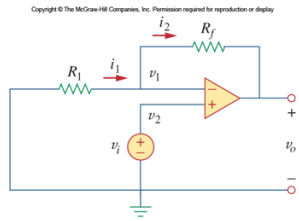


Equivalent circuit:



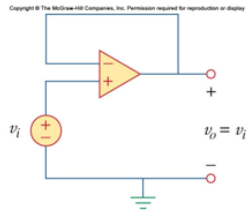
Non-inverting Op-Amp (Ideal)

- Ideal Op-Amp Rules apply
- Vo = Vi * [1 + (Rf/R1)]



Voltage Follower (Ideal, non-inverting op-amp)

Vi = Vo



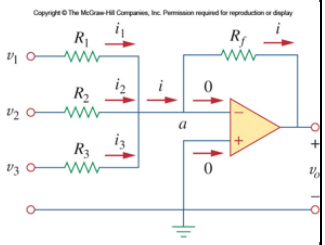
Summing Op-Amp

Uses the inverting amplifier and several inputs (each with their own resistor), the summing amplifier can be used to create a simple digital to analog converter (DAC).

I1 = [(V1-Va)/R1]
I2 = [(V2-Va)/R2]
I3 = [(V3-Va)/R3]

Ia = I1 + I2 + I3
Ia = [(Va-Vo)/Rf]

Vo = - [(Rf/R1)*V1 + (Rf/R2)*V2 + (Rf/R3)*V3]



Instrumentation Amplifier

-Places 2 non-inverting amplifiers before the difference amplifier to increase the impedance of the difference amplifier

$$v_o = \frac{R_2}{R_1}(v_{o2} - v_{o1})$$

$$v_{o1} - v_{o2} = i(R_3 + R_4 + R_3) = i(2R_3 + R$$

$$i = \frac{v_1 - v_2}{R_4}$$

$$v_o = \frac{R_2}{R_1}\left(1 + \frac{2R_3}{R_4}\right)(v_2 - v_1)$$

Difference Amplifier

-Vo is proportional to the difference between the two inputs.

-Va = Vb due to negative feedback.

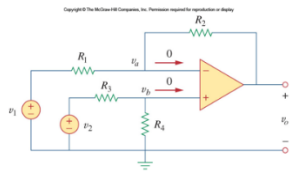
$$v_0 = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1$$

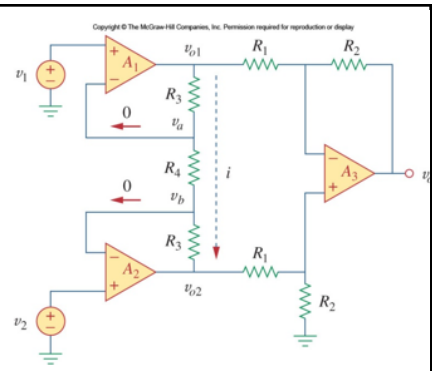
Common Mode Rejection: A difference amplifier rejects any signal that is common to the two inputs, following this equation:

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

If this condition is true for a difference op-amp, the output is then:

$$v_0 = \frac{R_2}{R_1}(v_2 - v_1)$$





Chapter 6 Concepts

Capacitors

- Capacitor Formulas:
  - Capacitance Formula: **C=(E*A)/d**
    - E = permittivity of dialectric
    - A = cross sectional area of plates
    - D = distance between plates
  - Charge stored in a capacitor is equal to the capacitance * Voltage: **q = C*V**
  - Take the derivative of this equation to find current in a capacitor: **I = C*(dv/dt)**
    - This is also the "ICE" part of the "ELI THE ICE MAN" acronym
  - To find voltage in a capacitor due to current:
    - $$v(t) = \frac{1}{C}\int_{t_0}^{t} i(\tau)\,d\tau + v(t_0)$$
    - The instantaneous power delivered to the capacitor is $p = vi = Cv\frac{dv}{dt}$
    - The energy stored in a capacitor is: $w = \frac{1}{2}Cv^2$
- Capacitor Properties:
  - When the voltage is NOT changing (constant), the current through the capacitor is zero (open circuit at DC conditions).
  - Voltage on the capacitor plates CAN NOT change instantaneously.
  - If the voltage on the capacitor does not equal the applied voltage, charge will flow until the cap reaches the applied voltage.
  - **Parallel Capacitors:** Act like series resistors, or like conductance values. C1 + C2 + C3 = Ceq
  - **Series Capacitors:** Act like Parallel Resistors, or like conductance values. [(1/C1) + (1/C2) + (1/C3)]^-1 = Ceq

Inductors

- Inductor formulas:
  - For a solenoid, the inductance formula is: L=(N^2*u*A)/l
    - U = permeability of the core material
    - N = number of turns of the wire
    - A = cross sectional area
    - L = length
  - If current is passed through an inductor, the voltage across it is proportional to the rate of change of current flowing through **L*(di/dt)**
    - This is the "ELI" part of the "ELI THE ICE MAN" acronym
  - The current stored in an inductor due to voltage:
    - $$I = \frac{1}{L}\int_{t_0}^{t} v(\tau)\,d\tau + i(t_0)$$
    - The power delivered to the inductor is: $p = vi = \left(L\frac{di}{dt}\right)i$
    - The energy stored is: $w = \frac{1}{2}Li^2$
- Inductor Properties:
  - If the current across an inductor is NOT changing (constant), th across an inductor is zero (short circuit at DC conditions).
  - The current through an inductor CAN NOT change instantanec
  - **Series Inductors:** Just like series resistors. L1 + L2 + L3 = Leq
  - **Parallel Inductors:** Just like parallel resistors. [(1/L1) + (1/L2) + (1/L3)]^-1 = Leq

Combining Op-amps, Capacitors, and inductors:

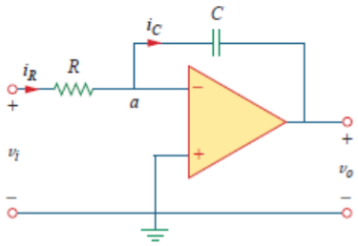
Integrator

Differentiator

Equation to find Vo:

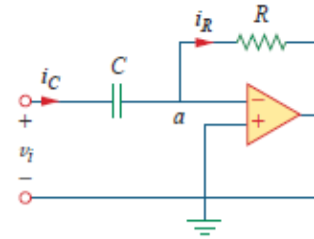
$$v_o = -\frac{1}{RC}\int_0^t v_i \, dt$$

$$i_R = \frac{v_i}{R}, \qquad i_C = -C\frac{dv_o}{dt}$$



$$i_R = -\frac{v_o}{R}, \qquad i_C = C\frac{dv_i}{dt}$$

$$v_o = -RC\frac{dv_i}{dt}$$



Chapter 7 Concepts:

First-Order Circuits (RC and RL Circuits)

- Circuit Excitation
  - No Independent Source (Natural Response)
    - a DC source is suddenly disconnected
    - Stored energy in the Capacitor/Inductor is then released to the system over a period of time
  - With independent Source (Forced Response)
    - DC, exponential, sinusoidal. Energy provided by sources
  - Natural Response + Forced Response = Complete Response

Source Free RC Circuits

- Since we can assume the capacitor was initially charged, at t=0 the initial voltage through the capacitor is:
  - V(0) = 0
- As the capacitor begins to discharge, the natural response starts to occur and voltage starts to dissipate:

$$v(t) = V_0 e^{-t/\tau}$$

  - Time constant (the speed at which the voltage drops to 1/e of the initial voltage):

$$\tau = RC$$

- With the voltage known, we can find the current:

$$i_R(t) = \frac{V_0}{R} e^{-t/\tau}$$

- The power dissipated in the resistor is:

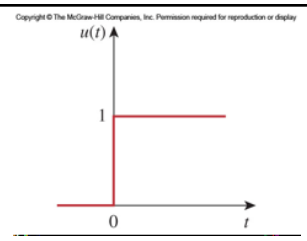
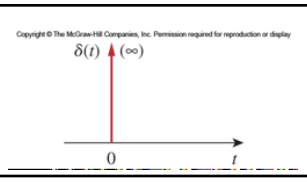
$$p(t) = \frac{V_0^2}{R} e^{-2t/\tau}$$

- The energy absorbed by the resistor is:

$$w_R(t) = \frac{1}{2} C V_0^2 \left(1 - e^{-2t/\tau}\right)$$

- The key to working with this type of situation is:
  - Start with the initial voltage across the capacitor and the time constant RC.
  - With these two items, the voltage as a function of time can be known.
  - From the voltage, the current can be known by using the resistance and Ohm's law.
  - The resistance of the circuit is often the Thevenin equivalent resistance.

Singularity Functions:

Unit Step Function $u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$ The switching time can be shifted to t=to by: $u(t - t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases}$	
Unit Impulse Function The derivative of the unit step function	

Step response of RC Circuits

- When a DC source is suddenly applied to a RC circuit, the source can be modeled as a step function.
- The complete response:

$$v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-t/\tau}, & t > 0 \end{cases}$$

  - The first part is the natural response of the capacitor/inductor due to the energy stored in it.
  - The second part is the forced response due to the connected voltage source. **Vs = V(∞)** and **Vo = V(0)**

- The complete response can be written as:

$$v = v_n + v_f$$

- Where the nature response is:

$$v_n = V_0 e^{-t/\tau}$$

- And the forced response is:

$$v_f = V_s\left(1 - e^{-t/\tau}\right)$$

- Note that the eventual response of the circuit is to reach V_s after the natural response decays to zero.
  - Once V(t) is found, current can be found:
  - i(t) =
    - $\frac{C}{\tau}\left[V(0) - V(\infty)\right] \cdot e^{-t/\tau}$
    - $= \frac{1}{R}\left[V(0) - V(\infty)\right] \cdot e^{-t/\tau}$

RL Circuits

- Source Free RL Circuits
  - Current cannot change instantaneously
  - We are looking for the current through the inductor
  - Therefore, we must determine its value as a function of time
    - Initial current passing through the inductor at t=0:
    - I(0) = 0
  - As the inductor begins to release energy into the system, the natural response starts to occur:

$$i(t) = I_0 e^{-Rt/L}$$

    - Time constant:

$$\tau = \frac{L}{R}$$

- Step Response of RL Circuits
  - Steady-state current through an inductor:

$$i_{ss} = \frac{V_s}{R}$$

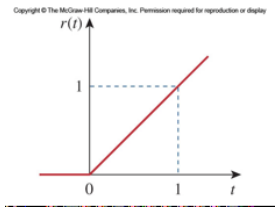
  - Current cannot change instantaneously through an inductor:
    - i(0+) = I(0-) = Io
  - Complete response of current through an inductor:

$$\delta(t) = \begin{cases} 0 & t < 0 \\ \text{Undefined} & t = 0 \\ 0 & t > 0 \end{cases}$$

Unit Ramp Function

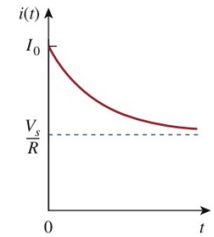
Integration of the Unit Step Function

$$r(t) = \begin{cases} 0, & t \le 0 \\ t, & t \ge 0 \end{cases}$$



$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right) e^{-t/\tau}$$

$$\tau = \frac{L}{R}$$



$$i(t) = i(\infty) + [i(0) - i(\infty)] \cdot e^{-t/\tau}$$

$$v(t) = L\frac{di}{dt} = \frac{L}{\tau}[i(0) - i(\infty)] \cdot e^{-t/\tau}$$

Chapter 8 Concepts

Second-order Circuits (RLC circuits)

- Start by getting initial conditions **I(0)** and **dI(0)/dt** (for parallel RLC circuits, **V(0)** and **dV(0)/dt**):
  - Capacitor: Open Circuit at long-term conditions. Voltage cannot change abruptly
  - V(0-) = V(0+) = Vo
  - Inductor: Short Circuit at long-term conditions. Current cannot change abruptly.
  - i(0-) = I(0+) = Io
- Figure out which damping case the circuit will require, solve characteristic eqn. for roots, apply them in correct formula, solve for I or V:
  - **Series RLC Circuits:**

▪ Overdamped (the roots are real and negative) $\alpha > \omega_0$	$i(t) = I_s + A_1 e^{\tau_1 t} + A_2 e^{\tau_2 t}$ (Overdampe
▪ Critically Damped (roots are real and equal) $\alpha = \omega_0$	$i(t) = I_s + (A_1 + A_2 t)e^{-\alpha t}$ (Critically Damp
▪ Underdamped (roots are complex) $\alpha < \omega_0$	$i(t) = I_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t)e^{-\alpha t}$ (Under

  - Damping factor, Undampened Natural Frequency, damped natural Frequency (shown below in order)

$$\alpha = \frac{R}{2L}, \qquad \omega_0 = \frac{1}{\sqrt{LC}} \qquad \omega_d = \sqrt{\omega_0^2 - \alpha^2}.$$

Parallel RLC Circuits:

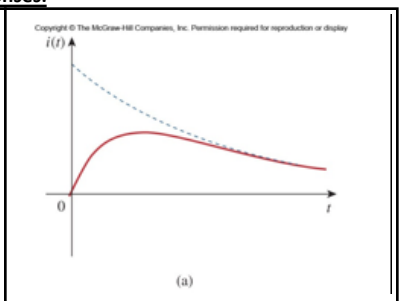
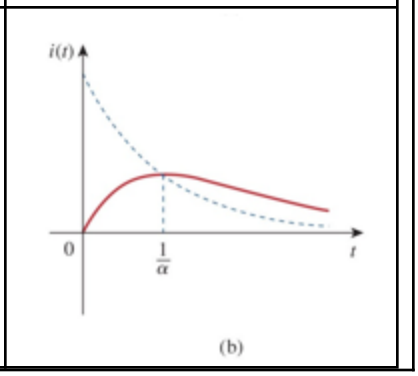
- All the same equations as series RLC, except:
  - Change I(t) --> V(t)

- $v(t) = V_S + A_1 e^{s_1 t} + A_2 e^{s_2 t}$ (Overdamped)
- $v(t) = V_S + (A_1 + A_2 t)e^{-\alpha t}$ (Critically Damped)
- $v(t) = V_S + (A_1 cos\omega_d t + A_2 sin\omega_d t)\, e^{-\alpha t}$ (Underdamped)

  - Vs = V(∞)
  - Change:

$$\alpha = \frac{R}{2L} \rightarrow \alpha = \frac{1}{2RC}$$

Damping Responses:

Overdamped	
Critically Damped	

- Solutions to characteristic equation:

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}, \qquad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

- Also can be found using quadratic formula:

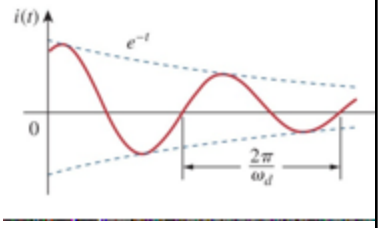
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- Underdamped case roots:

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -\alpha + j\omega_d$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -\alpha - j\omega_d$$

Underdamped



Chapter 9

- Sinusoidal Voltage: $V(t) = V_m \sin(\omega t \pm \phi)$
  - The function repeats every period, or every T seconds
  - $T = 2\pi/\omega$
  - Frequency (Hertz): $f = 1/T$
  - Angular frequency: $\omega = 2\pi f$
- Complex numbers:
  - Rectangular form: $z = x + jy$
  - Polar form: $Z = r\angle\phi$
  - Exponential form: $Z = re^{j\phi}$
    - Rectangular to polar:
      - $r = \text{sqrt}(x^2+y^2)$
      - $\phi = \tan^{-1}(y/x)$
    - Polar to rectangular:
      - $X = r\cos\phi$
      - $Y = r\sin\phi$
- Phasors
  - Converting from time domain to phasor:
    - $V(t) = V_m \cos(\omega t \pm \phi) \rightarrow V_m\angle\phi$
    - $V(t) = V_m \sin(\omega t \pm \phi) \rightarrow V_m\angle\phi - 90°$
    - Currents work the same way, just replace V_m with I_m.
  - Phasor Relationships (ELI THE ICE MAN)
    - Resistors: V=IR
      - Voltage and Current are in phase with each other (0° difference)
    - Inductors: $R = V/I = j\omega L$
      - ELI --> Voltage Leads Current by 90°, or current lags voltage by 90°
    - Capacitors: $R = -j/\omega C$
      - ICE --> Current leads voltage by 90°
  - Impedance --> phasor version of resistance
  - Admittance --> inverse of impedance, phasor version of conductance

Chapter 10

- Steps to analyze AC Circuits:
  - Transform the circuit to the phasor or freq domain
  - Solve the problem using circuit techniques
  - Transform back to time domain.
- Perform the following as in DC:
  - Nodal Analysis
  - Mesh Analysis
  - Superposition
  - Source Transformation
  - Thevenin and Norton Equivalents
  - Op Amp Analysis

- In the rectangular form of Impedance (z = x + jy), the real part (x) is the resistance, and the imaginary part is the reactance (jy)
  - When the impedance is positive, it's inductive (jwl), when it's negative it's capacitive (-j/ωC)
- In the rectangular form of admittance (z = x + jy), the real part (x) is the conductance, and the imaginary part is the susceptance (jy)

Chapter 11

- **Instantaneous Power**: $(1/2)V_mI_m[\cos(\phi_v - \phi_i)+\cos(2\omega t + \phi_v + \phi_i)]$
  - The rate at which an element absorbs power, or the power at any instant in time
  - 1st cosine is constant power, 2nd cosine is sinusoidal power
- **Average Power**:
  - $(1/2)V_mI_m\cos(\phi_v - \phi_i)$
  - ***Average power absorbed by an inductor and capacitor is zero watts.***
  - Average power is also:

For periodic current: $P = \frac{1}{T}\int_0^T p(t)dt$	$= \frac{1}{T}\int_0^T i^2 R dt$	$= \frac{R}{T}\int_0^T i^2 dt$
For DC Current: $P = i_{eff}^2 R$	$I_{eff} = \sqrt{\frac{1}{T}\int_0^T i^2 dt}$	$V_{eff} = \sqrt{\frac{1}{T}\int_0^T v^2 dt}$

- Resistive vs. Reactive Power
  - Resistive: When $\phi_v - \phi_i = 0°$, the voltage and current are in phase and the circuit is purely resistive
    - $P = (1/2)V_mI_m = (1/2)I_m^2R = (1/2)|I|^2R$
  - Reactive: When $\phi_v - \phi_i = \pm 90°$, the circuit absorbs no power and is purely reactive
    - $P = (1/2)V_mI_m\text{Cos}(\pm 90°) = 0$
- Maximum Average Power
  - $P_{max} = (|V_{Th}|^2)/(8R_{Th})$
- RMS
  - for a sinusoidal waveform, the RMS value is related to the amplitude as follows:

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

  - RMS Power can be determines from either RMS current or voltage:
    - $P = (I_{RMS})^2R = (V_{RMS})^2/R$
- **Complex Power**
  - $S = (1/2)VI^*$
    - I^* = complex conjugate of current
  - $S = P + jQ$
    - P = **Real power** --> $S\cos cos\left(\theta_v - \theta_i\right)$
    - jQ = **Reactive power** --> $S\sin sin\left(\theta_v - \theta_i\right)$
  - $S = V_{RMS}I_{RMS}^*$
  - $S = |V_{RMS}||I_{RMS}|<(\phi_v - \phi_i)$

$$\mathbf{S} = I_{\text{rms}}^2 \mathbf{Z} = \frac{V_{\text{rms}}^2}{\mathbf{Z}^*} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^*$$

- **<u>Apparent power</u>**
  - The product of RMS voltage and current will be called apparent power.
    - $|S| = |V_{RMS}||I_{RMS}| = \sqrt{P^2 + Q^2}$
- **<u>Power Factor</u>**
  - $P/S = \cos(\phi_v - \phi_i)$ --> $\cos^{-1}(P/S)$ = Power Factor Ratio (between 0 and 1)
    - P = Real Power
    - S = Apparent power
    - Real Power (P) = Apparent Power (S) * Power Factor (PF)
    - Power Factor = $\cos(\phi_v - \phi_i)$

- Adding a capacitor
  - To mitigate the inductive aspect of the load, a capacitor is added in parallel with the load.
  - With the same supplied voltage, the current draw is less by adding the capacitor.
  - $C = \frac{Q_c}{\omega V_{rms}^2} = \frac{P(\tan\theta_1 - \tan\theta_2)}{\omega V_{rms}^2}$